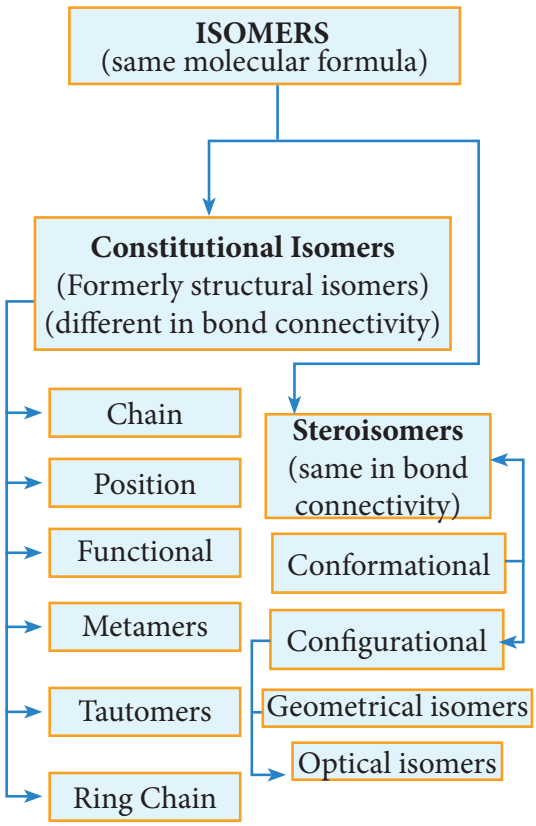
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of two or more compounds with the same molecular formula but different structure and properties (physical, chemical, or both). Compounds exhibiting this isomerism are called isomers. The difference in properties of two isomers is due to difference in (bond connectivity or spatial arrangement) the arrangement of atoms within their molecules. Isomerism is broadly divided into two types. i. Constitutional isomerism, ii. stereoisomerism.



11.5.1 Constitutional isomers (Formerly structural isomers):

This type of isomers have same molecular formula but differ in their bonding sequence. Structural or constitutional isomerism is further classified into following types.

(a) Chain or nuclear or skeletal isomerism:

These isomers differ in the way in which the carbon atoms are bonded to each other in a carbon chain or in other words isomers have similar molecular formula but differ in the nature of the carbon skeleton (ie. Straight or branched)

$CH_3—CH_2—CH_2—CH_2\text{-}CH_3$

n-Pentane

$CH_3\text{-}CH(CH_3)—CH_2\text{-}CH_3$

Isopentane
2-methyl butane

$CH_3—C(CH_3)_2—CH_3$

Neopentane
2,2-dimethyl propane

(b) Position isomerism:

If different compounds belonging to same homologous series with the same molecular formula and carbon skeleton, but differ in the position of substituent or functional group or an unsaturated linkage are said to exhibit position isomerism.

Example:

(i) Molecular formula C_5H_{10}

$CH_3—CH_2—CH_2—CH=CH_2$

Pent-1-ene

and

$CH_3—CH_2—CH=CH—CH_3$

Pent-2-ene